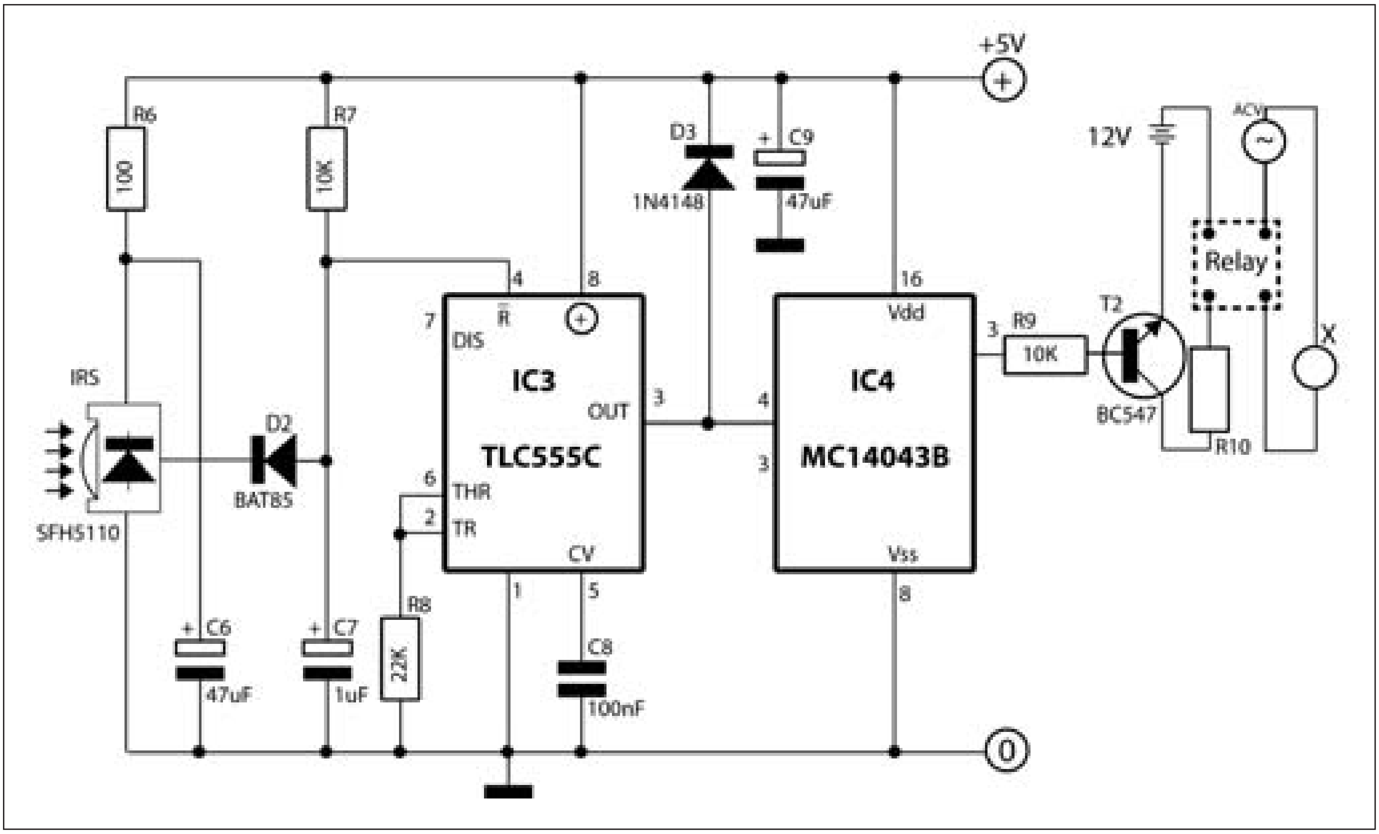


Receiver circuit diagram

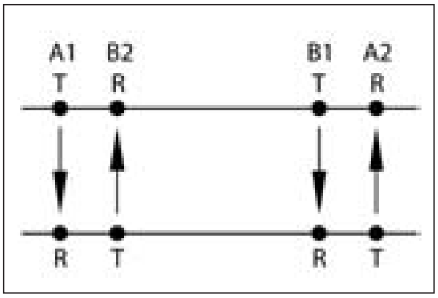
collector and the +5V. The IR LED draws power via the resistor R5, so if the distance between the receiver and transmitter is less, this can be reduced further to reduce the power consumption and allows the battery to last longer. Ensure the values of the components are very close to the mentioned values for the transmitter to emit the right frequency, this is of utmost importance.

The Receiver

The receiver uses another IC555 which obtains an input from the IR sensor (SFH5110) onto pin 4. You may substitute that with any of the following: SFH506-36, TFMS5360, TSOP1736, PIC12043S. Remember to cary P1 to generate the appropriate carrier frequency for the LED transmitter. When a signal of 36kHz is incident on the sensor, the sensor gives low digital output and when there is no signal of 36kHz, the sensor goes high. The diode D2 and capacitor C7 together eliminate the 300Hz modulation of the signal. So the sensor, when in a high state, sets the 555 to provide a high output which in turn switches the latch in IC4. The latch then sets the transistor T2(BC547) on. The BC547 acts as a switch for the relay which requires 12V to run on. The latch cannot provide a high voltage output which is why we use the transistor. The relay finally turns on your lights.

However, we have only turned on the lights so far. Now we need to turn them off. For this we have another receiver, except this time, the output of the 555 in the receiver is connected to pin 3 of the latch instead of pin 4. Pin 3 is the reset switch which will turn the relay off, thereby switching off your light.

Have 4 receiver transmitter combinations in the following arrangement to automate your lights completely.



Typical arrangement of sensors

A1 and B1 are connected to the same latch. A1 being connected to the Reset pin and B1 connected to the Set pin. A2 and B2 are also connected to another latch. A2 being Reset and B2 being the Set. When a person crosses B2, the lights turn on and switch off if either A1 or A2 are crossed. The same applies to people coming in from the other side.

This circuit though demonstrated for lights can be applied to any switch in the house. If you have pets in the house, then there will be a lot of accidental on/off cycles, so it'll be better if the transmitter/receiver pair are located at a height of 3 feet from the ground. Have fun and do write in.

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